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High-fidelity pixel-super-resolved lensless on-chip microscopy via height scanning and synthetic aperture

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Lensless on-chip microscopy (LOCM) has emerged as a promising imaging technique that combines a wide FOV with high-resolution capabilities. However, the imaging fidelity remains limited by persistent twin-image artifacts and the system complexity associated with illumination angle calibration. We present a high-fidelity, pixel-super-resolved (PSR) LOCM system that jointly integrates height scanning and synthetic aperture modulation, coupled with a phase retrieval algorithm designed for illumination separation. The system minimizes experimental complexity by eliminating the requirement for prior knowledge of the illumination angle and reducing axial translations. The proposed method achieves imaging resolution beyond the Nyquist–Shannon sampling limit, providing a 1.26-fold improvement in resolution across a full FOV of 28.6 mm². This method effectively suppresses twin-image artifacts, mitigates phase wrapping, and resolves fine structural details with superior fidelity. © 2025 Optica Publishing Group. All rights, including for text and data mining (TDM), Artificial Intelligence (AI) training, and similar technologies, are reserved.

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Lensless on-chip microscopy (LOCM) provides a distinctive advantage by achieving both a wide FOV and high resolution, effectively overcoming the spatial bandwidth product limitation inherent in conventional microscopes [1–3]. LOCM enables quantitative, label-free amplitude and phase imaging of complex samples, eliminating the reliance on external fluorescent dyes. Its lensless design removes the potential for optical and chromatic aberrations [4,5]. LOCM offers high-fidelity microscopic imaging solutions, which are particularly beneficial in the fields of physics and pathology [6–8].

The imaging resolution of LOCM is constrained by the pixel size of the image sensor, necessitating the integration of a pixel-super-resolved (PSR) phase retrieval algorithm alongside multiple measurement modes to enhance the system's spatial bandwidth product [9–11]. Typical modulation techniques include axial height scanning [12,13], wavelength scanning [1,4,14], multi-angle illumination [15–17], and encoded mask modulation [18,19]. The multi-frame alternating projection

(MFAP) phase retrieval algorithm incorporates a robust spatial support constraint and is typically integrated into LOCM [20,21]. Lensless imaging using a synthetic aperture (LISA) applies precise prior knowledge of extensive modulated illumination angles, significantly enhancing reconstruction accuracy [16,22]. To prevent high-frequency aliasing in the diffracted field, the distance between the sample and the image sensor should be extremely small [23]. This requirement proves challenging in axial height scanning systems due to stringent stability and precision demands.

In this Letter, we introduce a joint modulation LOCM that integrates height scanning and synthetic aperture, coupled with a height scanning synthetic aperture (HSSA) phase retrieval algorithm. Height scanning introduces diversity in the optical transfer function, thereby enhancing imaging quality. The synthetic aperture imaging scheme, based on illumination angle scanning, enhances the completeness of high-frequency information without relying on prior knowledge of illumination angles. Experimental validation on the USAF 1951 amplitude resolution target, an unstained chronic gastric ulcer slice, and a bee wing specimen demonstrates that the proposed method achieves PSR imaging beyond the Nyquist–Shannon sampling limit, providing a 1.26-fold improvement in resolution across a full FOV of 28.6 mm², effectively mitigating the inherent twin images.

The optical configuration is depicted in Fig. 1(a). When d_0 is significantly larger than d_n , the wavefront transmitted to the sample plane can be approximated as a monochromatic plane wave. In the experiments, d_0 is about 70 cm, and d_n is less than 2 mm. Axial scanning is conducted using a manual translation stage, which moves the sample along the optical axis across discrete positions. The synthetic aperture is implemented by shifting the pinhole. By manually repositioning the pinhole to different lateral offsets from the optical axis, oblique illumination angles are introduced. The illumination angles are randomly selected within a range of -8° to $+8^\circ$. We have provided a detailed analysis of the synthetic aperture in Supplement 1, Note 1. For various illumination directions, the hologram is projected onto different orientations of the CMOS sensor. No motorized or synchronized motion is required. These procedures are simple to implement and involve only basic mechanical operations. The

Algorithm 1. HSSA Algorithm

Input: Initialized $\varphi_0 = x_0 = \sqrt{I_1}$, $p_0 = \mathbf{1}^m$, and parameters $\alpha = 0.2$, $\beta = 0.7$, $\gamma = 0.05$, $T = 50$.
Output: Updated x_T

- 1: **for** $t = 1 : T$ iterations **do**
- 2: **for** $n = 1 : N$ Positions **do**
- 3: $S_n = D A_n T_n$
- 4: $y_{n,t} = S_n \varphi_t$
- 5: $\varphi'_{n,t} = S_n^{-1} [\sqrt{I_n} \cdot y_{n,t} / |y_{n,t}|]$
- 6: $\varphi'_t = \langle \varphi'_{1,t}, \varphi'_{2,t}, \dots, \varphi'_{N,t} \rangle$
- 7: $x_{t+1} = x_t + \frac{p_t^* \cdot (\varphi'_t - \varphi_t)}{(1-\alpha)|p_t|^2 + \alpha \max(|p_t|^2)}$
- 8: $p_{t+1} = p_t + \frac{\gamma x_t^* \cdot (\varphi'_t - \varphi_t)}{(1-\beta)|x_t|^2 + \beta \max(|x_t|^2)}$
- 9: $\varphi_{t+1} = p_{t+1} \cdot x_{t+1}$
- 10: **return** x_T

pre-processing steps for the experimental data are illustrated in Fig. 1(b). The low-resolution holograms are up-sampled, and the transform matrix D^{-1} is recorded. The ADFrFT autofocus algorithm [24] is then integrated to calculate the distance of each hologram from the sample, and the corresponding free-space light field propagation operator A_n^{-1} is computed. Finally, Fourier cross-correlation registration [25] is adopted to extract the sub-pixel displacements from each hologram, which does not require prior knowledge of the specific illumination angles or calibration of the illumination angle. In our data pre-processing, holograms are pre-propagated to the sample plane, enhancing their correlation and ensuring the reliability of the Fourier cross-correlation registration. The coordinate transformation matrix T_n^{-1} is recorded. For convenience, the matrices are vectorized. The intensity of the n th hologram is represented as $I_n \in \mathbb{R}^m$, where m is the total number of pixels in each hologram. The loss vector between the observed data and the retrieved result L is defined as follows:

$$L(x) = \frac{1}{2N} \sum_{n=1}^N \left\| |D A_n T_n(p \cdot x)| - \sqrt{I_n} \right\|_2^2, \quad (1)$$

where $x \in \mathbb{C}^m$ denotes the complex amplitude of the light field at the sample plane, $p \in \mathbb{C}^m$ denotes the process variable used to separate the background illumination wavefront, N denotes the number of frames captured by the CMOS sensor, and $\|\cdot\|_2$ represents the L_2 norm. The workflow of the HSSA phase retrieval algorithm for this non-convex optimization problem is illustrated in Fig. 1(c) and detailed in Algorithm 1. HSSA refines the free-space light field propagation matrix A_n into a more realistic form S_n to enhance imaging fidelity. Furthermore, HSSA employs a momentum-accelerated ptychography-based illumination separation strategy [26] that effectively suppresses twin images and environmental noise while maintaining reconstruction resolution. HSSA operates without requiring prior knowledge of the illumination angle, thereby reducing system complexity. The reconstructed complex amplitude and corresponding spectrum are illustrated in Fig. 1(d). The PSR spectrum exhibits an expanded range, with more prominent high-frequency components, confirming the imaging power spectral density enhancement. HSSA leverages multiple illumination angles to jointly constrain the reconstruction of a single complex amplitude at a specific axial height. This unified reconstruction strategy mitigates phase wrapping induced by abrupt variations in the optical transfer function.

Experiments were conducted using the USAF 1951 amplitude resolution target. The CMOS sensor (MV-GE1600C) used in the experiment has a pixel pitch of 1.34 μm . The raw hologram is shown in Fig. 2(a), and the corresponding full-FOV reconstruction is presented in Fig. 2(b). We use the binarized central region of the reconstructed USAF 1951 amplitude resolution target as the ground truth. Figures 2(c1)–2(c9) compare the reconstruction results of HSSA under varying numbers of axial heights and illumination angles, along with their corresponding mean squared errors (MSEs) relative to the ground truth. When the number of modulation heights is limited, the resolution is constrained by the singularity of the optical transfer function, leading to pronounced twin-image artifacts. When the input dataset comprises three axial heights, each associated with two illumination angles, HSSA resolves the 8th group, 6th line pair, and the MSE drops below 0.02 for the first time. Therefore, the minimum number of raw intensity measurements required is six.

Figure 3 presents a comparative analysis of the reconstruction results for the USAF 1951 amplitude resolution target. The comparisons among LISA, MFAP, and HSSA were all performed under the same number of holograms, with nine images in each case. LISA corresponds to the illumination angle scanning scheme, and MFAP corresponds to the height scanning scheme. The reconstruction results of LISA are shown in Figs. 3(a1) and 3(a2). Due to the interference of twin images, LISA reconstructions exhibit widespread artifacts, necessitating additional hologram inputs. The reconstruction results from MFAP are presented in Figs. 3(b1) and 3(b2). MFAP effectively suppresses artifacts, resolves the 8th group, 5th line pair (linewidth: 1.23 μm), and achieves pixel super-resolution beyond the Nyquist–Shannon sampling limit. The reconstruction results of HSSA are presented in Figs. 3(c1) and 3(c2), successfully resolving the 8th group, 6th line pair (linewidth: 1.10 μm), corresponding to a 1.26-fold improvement in resolution. Figures 3(a3)–3(c2) compare the amplitude profiles along the 7th group line pair, where HSSA reveals a more balanced and pronounced oscillation.

An experiment on the unstained chronic gastric ulcer slice was conducted. The LOCM platform is shown in Fig. 4(a). The phase sample and the corresponding imaging area are presented in Fig. 4(b). The full-FOV phase reconstruction result is illustrated in Fig. 4(c). To demonstrate the enhancement of imaging details achieved by HSSA, two regions of interest (ROIs) at different scales were selected for comparison. The reconstruction results of LISA and MFAP are presented in Figs. 4(d1) and 4(d2) and 4(e1) and 4(e2), respectively. The reconstruction results of HSSA are shown in Figs. 4(f1) and 4(f2), demonstrating more accurate phase retrieval and revealing finer structural details. To validate the accuracy of phase reconstruction, a simulation of phase reconstruction was performed in Supplement 1, Note 2.

The full-FOV reconstruction results of the bee wing specimen are presented in Fig. 5. The amplitude and phase reconstruction results of LISA shown in Figs. 5(a1) and 5(a2) exhibit pronounced artifacts. The amplitude and phase reconstruction results of MFAP shown in Figs. 5(b1) and 5(b2) effectively reconstructs fine structural details; however, residual twin-image artifacts are still apparent. Phase wrapping is illustrated in the magnified sub-region of Figs. 5(a2) and 5(b2). The amplitude and phase reconstruction results of HSSA are presented in Figs. 5(c1) and 5(c2), respectively. Fine structural features, including hairs on the wing veins and membranes, are clearly

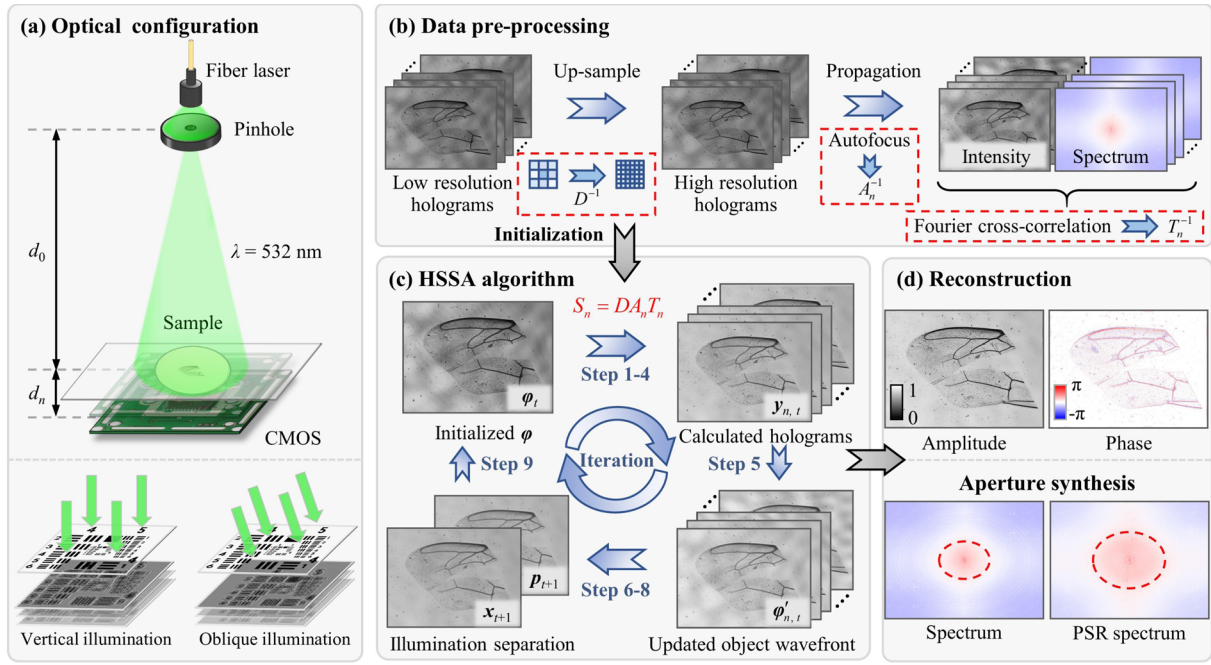


Fig. 1. Schematic diagram of the LOCM system and HSSA algorithm.

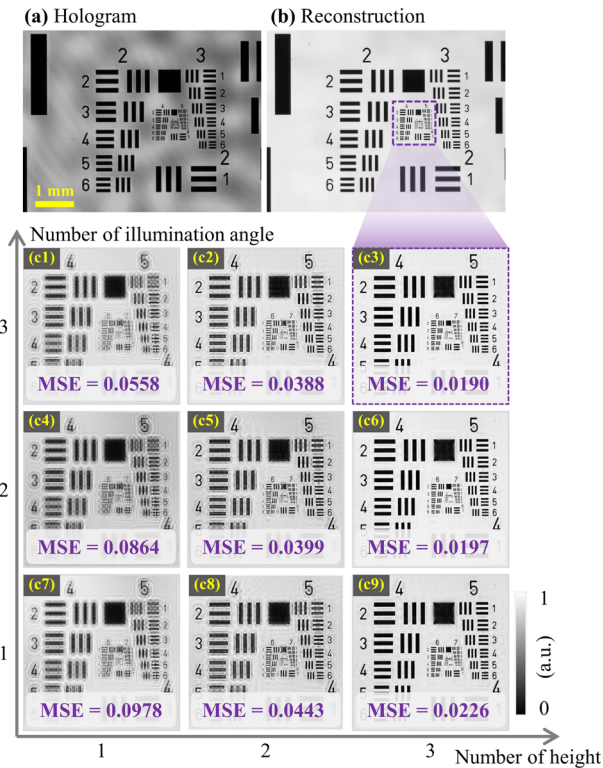


Fig. 2. Comparative experimental reconstructions of the USAF 1951 amplitude resolution target. (a) Raw hologram. (b) Full-FOV amplitude reconstruction result. (c1)–(c9) Reconstruction results with varying axial heights and illumination angles.

resolved, with twin-image artifacts significantly suppressed. In terms of phase reconstruction, HSSA effectively mitigates phase wrapping and demonstrates excellent imaging performance for complex, thick samples. To further verify the robustness of

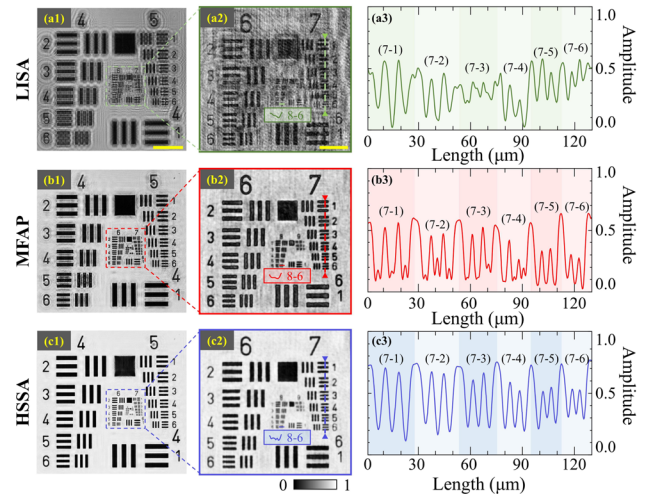


Fig. 3. Experimental reconstruction of the USAF 1951 amplitude resolution target. (a1)–(c1) Amplitude reconstruction results of LISA, MFAP, and DCNS. (a2)–(c2) Magnified views of the central regions in (a1)–(c1), respectively. (a3)–(c3) Amplitude profiles along the 7th group corresponding to (a2)–(c2). Yellow scale bar: 200 μ m in (a1)–(c1) and 50 μ m in (a2)–(c2).

HSSA in pixel-super-resolved imaging, the experimental reconstruction result of the pine stem cross section is conducted in [Supplement 1, Note 3](#). HSSA reconstructs the intricate structural details, providing additional evidence of its high fidelity and stability. Time required for image sequence capture and reconstruction is analyzed in [Supplement 1, Note 4](#). HSSA offers improved computational efficiency along with superior imaging fidelity.

In summary, we propose a joint modulation LOCM scheme that integrates height scanning and synthetic aperture, coupled with an HSSA phase retrieval algorithm specifically designed

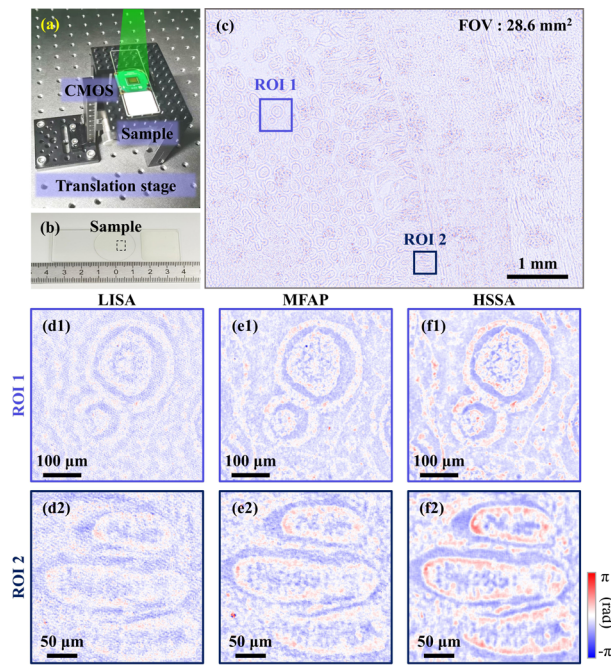


Fig. 4. Experimental reconstruction of the unstained chronic gastric ulcer section. (a) LOCM platform. (b) Phase sample and its corresponding imaging area. (c) Full-FOV phase reconstruction result. (d1)–(f1) Phase reconstruction results of ROI 1 of LISA, MFAP, and HSSA, respectively. (d2)–(f2) Phase reconstruction results of ROI 2 of LISA, MFAP, and HSSA, respectively.

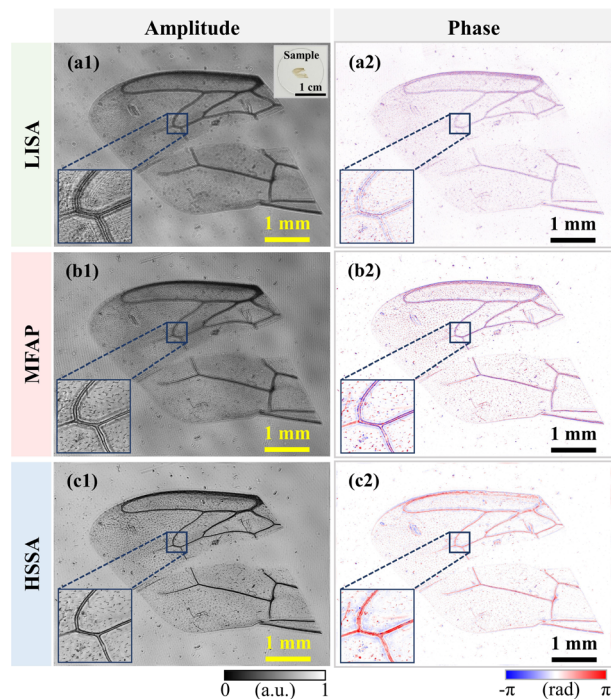


Fig. 5. Experimental reconstruction of the bee wing specimen. (a1)–(c1) Amplitude reconstruction results of LISA, MFAP, and HSSA, respectively. (a2)–(c2) Phase reconstruction results of LISA, MFAP, and HSSA, respectively.

for the experimental configuration. The proposed method eliminates the need for determining specific illumination angles or performing precise calibration of the experimental setup. By expanding the illumination numerical aperture, the effective numerical aperture of the imaging system is increased, ensuring the completeness of high-frequency information. Without complex post-processing, HSSA achieves higher resolution with a limited number of measurements. Twin-image artifacts are significantly suppressed, and phase wrapping is effectively mitigated. HSSA demonstrates superior imaging performance for complex samples and offers a reliable and promising tool for biomedical diagnostics. For future work, we will focus on the limits of high-frequency information recovery under pixel binning and sampling constraints and explore acceleration strategies for the reconstruction algorithm to further enhance the resolution and efficiency of the LOCM system.

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Data availability. The data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

Supplemental document. See Supplement 1 for supporting content.

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